

2. Three kinds of “model” (why you got lost)

Last synced: 2026-07-28

Audience: Charles, re-entering

Standalone: definitions are inline; no other files required.

Slide summary (same story): [Model.pdf](#)

The mistake that feels like failure but is not

You kept hearing “the model” and assumed one equation should do everything: fit the hero, explain NBA psychology, simulate a league, and satisfy Alex — all at once.

This project actually uses **three different objects**. They stack, but they are **not the same thing**. Confusion came from switching layers mid-sentence — not from you being unable to do math.

BINDING insight — environment ≠ advancement; score ≠ select (#1 confusion)

Charles lock (Jul 2026; sharpened Jul 2026). Full binding doc: [../BINDING_Selection_is_its_own_step.md](#).

You were merging mechanisms that must stay separate:

- Environment / development** — peers affect you through benefit (B) and congestion (D). The **combined environment object** is $L_{net} = B - D$. This **describes the peer environment**, not who wins the slot.
- Advancement** — who gets the scarce slot — itself has **two** pipeline steps:
 - Scoring** — rank candidates: $S_i = A_i - \lambda \cdot L_C$ (λ lives here).
 - Selection (winner rule)** — turn ranks into winners: **top K** in v1 (later: soft / stochastic draw).

	Environment ($L_{net} = B - D$)	Scoring (S_i)	Selection (winner rule)
Asks	How do peers help vs hurt development?	How do we rank ?	Given ranks, who wins ?
Knob	(theory; not estimated in v1)	λ	Top K now; noise later
Hero plot	Does not separate channels	Shows outcome only	—
Sim	Who lands on which roster	Build S_i	Draft by winner rule

Carry this sentence: *The hero describes outcomes; L_{net} is the peer environment; advancement = **score** then **select**; the sim tests whether the **weight λ on L_C in the score** changes who gets selected under a fixed winner rule.*

Layer C is powerful because it **names score and select as steps** — talent-only score vs congestion-in-score, same top-K — instead of hiding advancement inside a giant environment model.

Layer A — describe the curve (phenomenology)

Job: Answer “What does the data look like?”

Tools: Bin the real panel. Optionally fit a simple curve (e.g. draft probability vs pool quality plus its square).

Basketball example: The hero plot and a quadratic regression on the same filtered rows.

We fit a **quadratic** (not a line) because the hero bends at the top — it is the simplest way to ask whether draft rate is **concave** in pool quality (negative squared term) and to draw a smooth overlay on the bins. We are **not** claiming NBA teams use that formula; the **binned hero** stays the headline stylized fact.

Why the quadratic LPM is still useful (brief): Fit $Y \approx \beta_0 + \beta_1 Q + \beta_2 Q^2$ on the same hero rows (LPM = linear probability model = OLS on a 0/1 draft outcome). Then:

- $\beta_2 < 0 \Rightarrow$ fitted curve is **concave down** (inverted-U / hill); $\beta_2 > 0 \Rightarrow$ **concave up** (U).
- One turning point at $Q^* = -\beta_1 / (2\beta_2)$ — a **local max** if $\beta_2 < 0$, a **local min** if $\beta_2 > 0$.
- A pure quadratic has **no inflection point** (second derivative is constant). Do not claim inflection from this fit.
- Speak about the **fitted quadratic**, not “the data are definitively a parabola.” Fitted values can leave $[0, 1]$ — fine for **shape**, not a full probability model. Mechanism remains Layer B/C.

What success looks like: Negative curvature — draft rate bends down at the top bins — on the **locked** hero definition (16 bins, leave-one-out quality, minutes filter, seasons 2011–2021).

What this does NOT prove:

- Why the curve exists.
- That NBA teams use our formula.
- That simulation must match bin-by-bin.

Analogy: Layer A is a **weather report**. It records the shape of the cloud. It is not a **theory of rain**.

Layer B — explain help vs hurt (mechanism story)

Job: Answer “Why could good peers both help and hurt?” — and **separate environment from advancement** (and, inside advancement, **score from select**).

Part 1 — Environment (B minus D)

Net value from the peer environment splits into two parts that pull in opposite directions:

- **B (benefit)** — upside from good peers: development, visibility, playing in a strong context.
- **D (congestion / dilution)** — downside from good peers: harder to stand out, more competition around you.

Write this as $L_{net} \approx B - D$ (net local environment). As pool quality rises, both B and D can increase. Early on, benefit may dominate (draft rate rises). At the very top, congestion can catch up or outweigh benefit — a story for why the hero tail might dip. In v1 we are not estimating separate B and D curves from data.

This part describes the environment — not who gets drafted.

Part 2 — Advancement: score, then select

Advancement is separate from environment. After rosters exist, someone must decide who gets the scarce slot. That happens in two steps:

1. **Scoring** — compute a rank for each player.
2. **Selection (winner rule)** — pick winners from those ranks (v1: top K; later: soft / stochastic draw).

Alex’s equation is about scoring, not the full environment and not the winner rule:

- $S_i = A_i - \lambda \cdot L_C$ — own ability minus weight x viable-peer congestion in the score (equivalently $S_i = A_i + \lambda(B-D)$ with $(B-D) = -L_C$ in the score for Alex v1)
- L_C — D-ish congestion measured LOO on a roster after assignment; not full L_{net} , not $L_C = f(B, D, \rho)$ in v1
- λ — weight on congestion in the score only

That is why the simulation makes advancement explicit: assign → score → select (top K). Knockout (repo shorthand — say mechanism contrast to Alex if you prefer): hold the fake league and the winner rule fixed; remove congestion from the score. Compare top-K on $S_i = A_i$ only ($\lambda = 0$) vs top-K on $S_i = A_i - \lambda \cdot L_C$. The one ingredient toggled is L_C in the score — not talent, not top-K, not the whole L_{net} environment.

What success looks like: The story is coherent; sim shows ability-only scoring fails the congestion-in-score story when $\lambda = 0$.

What this does NOT prove in v1:

- Separate measurement of B(Q) and D(Q) on one axis.
- That real NBA front offices compute Alex’s score literally.
- That the hero curve proves the scoring channel only (hero is outcome; channel is theory + sim).

In plain terms: Layer B Part 1 = environment (B – D). Layer B Part 2 = score (S_i) then select (top K). Layer C = run those steps in code.

Unified nesting — how the pieces fit (beyond v1)

One way to nest the symbols without merging layers:

General score: $S_i = A_i + \lambda \cdot (B - D)$ (ability plus weighted net local environment in the ranking).

Restriction	Meaning	Score
Knockout	$\lambda = 0$ — remove all of (B – D) from the score	$S_i = A_i$
Alex v1	Only congestion enters the score: (B – D) = $-L_C$ (not “B is zero in the world”)	$S_i = A_i - \lambda \cdot L_C$
Full (later)	$\lambda \neq 0$ with both B and D in (B – D)	Richer; not what v1 runs

Full generative pipeline (Layer C — Charles’s three steps):

Draw A_i → assign to rosters (ρ) → compute LOO poolq_loo, L_C on rosters → score: $S_i = A_i + \lambda(B-D)$ [v1: (B-D)=- L_C] → select: top K [later: stochastic winner rule] → bin for plots
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Knob	Step	Role
ρ (rho)	Assignment (who lands where)	Assortativity in soft match to T_{ij} ; $\rho=0$ = max mixing; $\rho\uparrow$ = sharper match; not inside S_i
λ	Scoring	Weight on (B – D) in S_i ; v1 knocks out via $\lambda = 0$ or sets (B – D) = $-L_C$
Top K / noise	Selection (winner rule)	How ranks become winners; v1 = deterministic top K
B, D	Environment (theory)	$L_{net} = B - D$ — help vs hurt among peers; not separately estimated in v1
L_C	Computed on rosters after assignment	D-ish operational congestion (LOO viable-peer crowding, e.g. crowding_smooth) — not the same object as full L_{net} , and not a clean formula $L_C = f(B, D, \rho)$

Minimal generative claim (Alex v1): some roster step (so LOO stats exist) and score → select (congestion in vs out of S_i , same top-K). Calibrating ρ to 530 or matching hero bins is not the v1 gate.

Hero mapping (stay honest): The hero is **Layer A outcome** — mean draft rate by `poolq_loo` on the **real panel**. It **motivates** the story; it does **not** identify (ρ , λ , B , D) or prove `S_i` is the NBA rule. `poolq_loo` is a **pool proxy**, not measured ($B - D$).

`L_C` and θ (**viability cutline**): On a roster, `L_C` = LOO congestion from **viable peers**. θ = viability threshold (530 default: `med(perf | ever drafted)`). **Hard**: count teammates with `A_j > \theta`. **Smooth (default)**: LOO mean of `\sigma(\gamma(A_j - \theta))` (`crowding_smooth`). γ = sharpness of the sigmoid around θ .

Optional follow-up (Alex asked Jul 2026): ρ sim ablations — fix **score and winner rule**, vary assignment assortativity — test "is sorting involved?" **Not** required for v1 minimal model; the headline knockout is $\lambda = 0$ vs $(B-D)=-L_C$ in `S_i` (already run; see doc 03).

Layer C — simulate a fake league (generative proof-of-concept)

Job: Answer “If we write explicit draft rules, does congestion in the score change who gets picked?”

Tools: Thin `540_*` / scripts (archived 538D is reference only): create synthetic players → assign to teams → **score** → **select** (top K) → plot draft rate by bins.

Headline comparison (knockout on `L_C` in the score; winner rule fixed):

- **Knockout arm ($\lambda = 0$):** `S_i = A_i` — congestion **removed** from the **score**.
- **Full rule for this test:** `S_i = A_i - \lambda \cdot L_C` — same league, same top-K, congestion in the score.

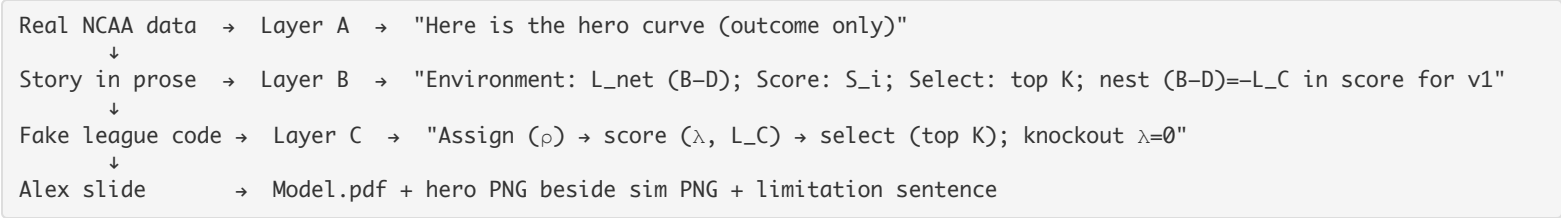
What success looks like: The two **scoring** rules produce **different shapes** under the same winner rule — especially, talent-only scoring fails to show the “congestion story.”

What this does NOT prove:

- Point-for-point match to every hero bin.
- That the real draft is this simulation.

Analogy: Layer C is a **wind-tunnel model**. It shows a mechanism **can** produce bent curves. It is not claiming the wind tunnel **is** the sky.

How the three layers stack (one picture in words)



You do **Layer A** first (mostly done). You need **Layer C minimal** for the simplified model deadline. **Layer B** has environment ($B - D$) plus **score then select** — you are not failing if you have not “estimated Layer B” as one formula.

What Alex side-by-side actually means (v1)

Not: “Sim reproduces hero exactly.”

Yes:

- Same **kind** of Y-axis (advancement / draft rate).
- **Honest** X-axis labels (empirical leave-one-out quality vs sim bins on the best matching axis you have).
- One clear **limitation sentence**: we show congestion in the **score** can bend who gets selected under top-K; we do **not** claim bin-for-bin replication yet.

That is enough for the simplified model chapter of work.

Notation landmines (same symbol, different day)

While re-entering, ignore cross-domain notation. For **basketball v1 only**, use this cheat sheet:

Symbol / term	Meaning here	Ignore for now
<code>L_net</code>	Net peer environment = B – D (help minus hurt among peers)	Any other “L” in old emails
<code>S_i</code>	Score (Alex) — ranking for advancement; not the winner rule	Hero regression coefficients
Selection	Winner rule — top K now; later stochastic draw from scores	Calling <code>S_i</code> “selection”
<code>L_C</code>	LOO viable-peer congestion on a roster (used in <code>S_i</code>); D-ish, not full <code>L_net</code>	Network centrality papers
θ (theta)	Viability outline for “substitutable peer” — e.g. med(A drafted)	Generic threshold in other domains
γ (gamma)	Sharpness of <code>o(γ(A_j – θ))</code> in smooth <code>L_C</code>	—
ρ (rho)	Assignment assortativity — soft match to <code>**T_{j*}</code> ; $\rho=0$ = max mixing ; not inside <code>S_i</code> **	Legacy τ (temperature) in archived docs (opposite intuition)
Knockout	Same league + same winner rule; <code>L_C</code> removed from the score ($\lambda = 0$) vs Alex score with it in	“Mechanism contrast” to Alex
BINDING	<code>L_net</code> $\neq$ advancement; score $\neq$ select ; hero $\neq$ scoring equation	—
Pool quality (LOO)	Leave-one-out teammate quality	Army “poolq” without LOO
Ability (<code>A_i</code>)	Player talent in sim / own perf in data	Tenure “productivity”
λ (lambda)	Weight on <code>L_C</code> in score: <code>S_i = A_i - \lambda L_C</code> (λ is not <code>L_C</code> itself)	—
K	Selection capacity — count of winners (top-K); code: <code>n_selected</code>	Old memos used Λ for this — retired
K/N	Selectivity rate (system feature); characterization default 10%	MBB draft ~1% is a domain point, not the general baseline
Hero	Empirical binned draft plot	Any other “hero” in old emails

When a doc uses shorthand without re-defining, **close it** and come back to this table.

Notation addendum (Aug 2026) — read before Phase B

Purpose: One page so λ , K, and K/N stop swapping roles. Full threshold notes: `06_Lambda_threshold_and_KN_memo.md` .

Score vs select (BINDING)

Step	Symbol	Plain English	Code
Score	λ	Weight on congestion in the score — how much <code>L_C</code> is subtracted from <code>A_i</code>	<code>loo_gap_weight</code>
	<code>L_C</code>	Viable-peer congestion (built with θ , γ)	<code>pool_c_smooth_loo</code>
	<code>S_i</code>	Ranking number: <code>S_i = A_i - \lambda L_C</code>	<code>selection_weight</code>
Select	K	How many get Y = 1 (top-K)	<code>n_selected</code>
	K/N	Selectivity rate — system feature	<code>K / (n_teams × roster_size)</code>

λ is not “the congestion.” Congestion is `L_C`. λ is only the **weight** on that term in `S`.

K is not λ . K counts winners after scoring. Old memos used **Λ** for K — retired.

Characterization defaults (gallery)

Preset	N	K	K/N	When to use
<code>characterization</code> (default)	5600	560	10%	Phase B knob sweeps
<code>mbb_draft</code>	5600	56	~1%	Domain calibration (later)
<code>army_high</code>	5600	2240	40%	High-selectivity regime (explore)

Set via `GALLERY_K_OVER_N` or `GALLERY_N_SELECTED` (see `sports/scripts/gallery_knobs.py`).

θ — do not set by rule until tested

θ = center of $\sigma(\gamma(A-\theta))$ when building `L_C` (who counts as a “viable peer”).

- 539 sim default:** $\theta = 0.72$ on [0,1] Beta **A** (Alex reference JSON).
- 530 empirical:** $\theta \approx \text{med}(\text{perf} \mid \text{ever drafted})$ on ppm **z** — different scale.

Open (PD15): Does θ co-vary with K/N ? Run $\theta \times K/N$ panel *before* adopting any fixed rule for θ . Script:

`sports/scripts/theta_kn_sweep_diagnostic.py` .

ν , ρ (unchanged)

- ν — sharpness of viability sigmoid (not assignment).
- ρ — assignment assortativity (not inside S_i).

λ_{crit} (sort-and-chop benchmark)

Critical λ = first λ where **score ranking** ($S_i = A_i - \lambda L_C$) differs from **talent-only ranking** — not “when congestion feels big.” Full derivation and the **θ -straddle team** picture: `06_Lambda_threshold_and_KN_memo.md` § Critical λ , § The θ -straddle team.

What to tell yourself when reading old memos

If a one-page memo assumes you already know which layer is which, it will feel like gibberish. That is a **format mismatch**, not missing intelligence.

While lost: read only re_entry docs 01–03.

When steady: optional short memos become useful again.

Next: [03_Three_Day_Basketball_Focus.md](#)